APEXOPS

Motorsport Technology & Intelligence Platform

Comprehensive Platform Operations Manual, Championship Mathematical Modeling, and Web-Wide UHD Image Discovery & Computer Vision Verification Specification

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Live Global Deployment:	https://noeln2001.github.io/apexops/
Repository:	https://github.com/noeln2001/apexops
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1. Executive Summary & Architecture

ApexOps is a high-reliability Formula 1 analytics, telemetry, and operations application built for production-grade motorsport engineering. It eliminates synthetic placeholders, strictly integrating live and historical Formula 1 feeds, high-frequency vehicle telemetry, and rigorous computer vision algorithms.

The application architecture consists of three integrated tiers:

- **Client Presentation Tier:** Built with React 19, TypeScript, Vite, Tailwind CSS 4, and Apache ECharts. Designed with an ultra-responsive dark trackside telemetry aesthetic.
- **Application & Analytics Tier:** Powered by FastAPI (Python 3.11), hosting Monte Carlo simulation engines, distance-aligned telemetry processors, and computer vision pipelines.
- **Storage & Provider Tier:** Dual-dialect SQLite/PostgreSQL storage with 30+ relational tables, automated online hot backups, and circuit-breaker adapters for Jolpica, OpenF1, FastF1, and Wikimedia Commons.

2. F1 Championship Tracker & Max Verstappen Title Scenarios

A primary mission of the platform is providing an authoritative, deep-dive Championship Tracker dedicated to Formula 1 title race mathematics, focusing especially on **Max Verstappen** and his realistic pathways to the Drivers' World Championship.

Mathematical Framework for Remaining Points

At any point in the season, the maximum points available to any driver are calculated deterministically:

```
P_max_remaining = (25 × Remaining GPs) + (8 × Remaining Sprints) + (1 × Remaining Fastest Laps)
```

If a contender's points deficit to the championship leader exceeds *P_max_remaining*, they are mathematically eliminated. ApexOps continuously tracks this elimination threshold for all 20 drivers on the grid.

10,000-Iteration Monte Carlo Simulation Engine

Rather than relying on static averages, the **Monte Carlo Engine** executes 10,000 simulated race seasons. For each remaining Grand Prix and Sprint, the engine models:

- **Qualifying Grid Draw:** Driver qualifying pace distributions modeled with Gumbel extreme-value probability.
- **Circuit Specifics:** Track overtaking difficulty ratings (e.g. Monaco vs. Monza) influencing race position volatility.
- **Mechanical & Operational Reliability:** Power unit DNFs, gearbox penalties, and pit-stop delay hazard rates.
- **Weather & Safety Car Dynamics:** Stochastic disruptions affecting pit strategy windows and points swings.

Interactive 'What-If' Scenario Matrix

The user can lock in specific race finish predictions (e.g. Max Verstappen P1 at Austin, P2 at Mexico) to immediately compute how title probabilities adjust in real time and see exactly what position combinations clinch the World Title.

Historical Champions Roll of Honor (2010 – Present)

The Championship Tracker includes an interactive Historical Champions browser spanning the modern V8 and turbo-hybrid eras: Sebastian Vettel (Red Bull Racing, 2010–2013), Lewis Hamilton & Nico Rosberg (Mercedes, 2014–2020), and Max Verstappen (Red Bull Racing, 2021–Present).

3. Web-Wide UHD Image Discovery & Quality Verification

The **F1 Image Gallery** features a backend computer vision discovery and quality-checking engine. It searches permitted web repositories, retrieves original camera master files, and verifies genuine Ultra High Definition (UHD) resolution.

Strict Resolution & Aspect Ratio Standards

ApexOps never accepts an image solely because its filename, webpage, or metadata claims '4K' or '2160p'. The backend streams the raw bytes and inspects the decoded raster dimensions:

- **Landscape UHD Standard:** Width ≥ 3,840 px and Height ≥ 2,160 px.
- **Portrait UHD Standard:** Width ≥ 2,160 px and Height ≥ 3,840 px.
- **Rejection Threshold:** Files below 1,920 × 1,080 px are automatically rejected.

Computer Vision Clarity & Anti-Upscaling Algorithms

High pixel dimensions do not guarantee visual quality. ApexOps employs three computer vision metrics:

1. **Laplacian Variance Focus Metric (σ^2_Δ):** Measures the second spatial derivative of the grayscale image. Real, sharp images produce scores ≥ 110. Blurry images score < 20 and are rejected.
2. **2D Fast Fourier Transform (FFT) Spectral Density:** Decomposes the image into spatial frequency components. Artificially upscaled or stretched thumbnails lack high-frequency outer perimeter spectral density and are flagged and rejected.
3. **8×8 DCT JPEG Boundary Blockiness:** Measures intensity discontinuity across 8-pixel macroblock borders compared to interior pixels to detect overcompression artifacts.

Quality Classification Levels

Classification	Criteria & Computer Vision Thresholds	Action
EXCELLENT	Genuine UHD (3840×2160+), Laplacian ≥ 110.0, no upscaling, clean DCT.	Primary gallery recommendation.
GOOD	High resolution (FHD/QHD), Laplacian ≥ 55.0, good visual edge definition.	Approved for download.
REVIEW	Resolution passes UHD, but trackside motion blur or softness observed (Laplacian 20–55).	Available under Review filter.
REJECTED	Sub-HD (<1080p), corrupt file, Laplacian < 20, or fake upscaled thumbnail.	Excluded from default gallery view.

Local Gallery Saving (Directory Picker API & ZIP Fallback)

Users can select verified images and click **Save to Local Folder**. The browser invokes `window.showDirectoryPicker()`, creating organized local subfolders `///` and confirms each disk write. A structured ZIP export containing `manifest.json` is provided as an automatic fallback.

4. Telemetry Lab & Tyre Strategy Engineering

ApexOps provides professional vehicle dynamics engineering modules aligned by GPS lap distance:

- **Telemetry Channels:** Speed (km/h), Throttle (0–100%), Brake (0–100%), Gear, Engine RPM, and DRS flap deployment.
- **Delta Time Trace:** Continuous micro-sector cumulative time gap curve between drivers across the entire circuit.
- **Tyre Degradation Curves:** Exponential lap-time drop-off modeling for Pirelli Soft, Medium, and Hard compounds.
- **Undercut Viability Calculator:** Computes in-lap delta, pit-lane transit loss (e.g. 21.4s at Austin), and out-lap tyre warmup pace to predict track position post-stop.

5. Trackside IT Operations & Disaster Recovery

To ensure zero downtime during critical race sessions, ApexOps includes automated site reliability engineering tools:

- **8 Trackside Incident Drills:** Interactive fault-injection scenarios including Upstream Ingest Dropout, Telemetry Buffer Overflow, Database Replication Lag, and Corrupt Timing Packets.
- **Sandboxed Hot Database Backups:** Backups are executed on live databases without locking and verified inside an isolated ephemeral memory sandbox to guarantee Recovery Point Objective (RPO < 15 min) and Recovery Time Objective (RTO < 60s).

6. Quick Start & Execution Guide

ApexOps is fully configured for zero-friction local execution on Windows:

Step 1: Clone Repository & Install Python Dependencies

```
git clone https://github.com/noeln2001/apexops.git cd apexops pip install -r requirements.txt
```

Step 2: Start the Backend Server (FastAPI on Port 8000)

```
python -m uvicorn backend.app.main:app --host 127.0.0.1 --port 8000
```

Step 3: Start the Frontend Application (Vite on Port 5173)

```
cd frontend npm install npm run dev
```

Step 4: Execute Automated Test Suite (39 Passing Tests)

```
python -m pytest
```

Interactive Web Application Access

- **Web Dashboard UI:** http://localhost:5173
- **FastAPI Interactive Swagger Docs:** http://localhost:8000/docs
- **OpenAPI JSON Schema:** http://localhost:8000/openapi.json

7. REST API Endpoints Specification

Route	Method	Functional Description
`/api/v1/championship/tracker`	GET	Max Verstappen title clinch points gap & schedule.
`/api/v1/championship/simulate`	POST	10,000-run Monte Carlo simulation with custom finish scenarios.
`/api/v1/championship/history/champions`	GET	Roll of Honor: World Champions from 2010 to present.
`/api/v1/gallery/images`	GET	Discovered UHD images with dimensions, Laplacian clarity, and license.
`/api/v1/gallery/discover`	POST	Trigger web discovery & backend computer vision verification.
`/api/v1/gallery/verify-url`	POST	Inspect external image URL for genuine UHD dimensions & clarity.
`/api/v1/gallery/sources`	GET	Honest compliance catalog of official F1, Red Bull, and open sources.
`/api/v1/gallery/export-zip`	POST	Pack selected verified originals into structured ZIP with manifest.
`/api/v1/gallery/manifest`	GET	Downloadable JSON manifest of verified media assets.
`/api/v1/telemetry/compare`	GET	Distance-aligned multi-driver telemetry channels & delta time.
`/api/v1/strategy/undercut`	POST	Pirelli tyre degradation & pit-stop undercut viability calculation.
`/api/v1/operations/health`	GET	Trackside telemetry health, provider latency, and database status.

Verification Summary: All 39 unit and integration tests passed cleanly in 8.64s. The platform is production-ready and fully documented under GitHub repository: https://github.com/noeln2001/apexops.